

The American Academy of Neurology (AAN) recommends EEG as one of the ancillary methods used to confirm cerebral death in combination with a standardized clinical evaluation [1]. The EEG when recorded using the minimal technical standards as outlined by the American Clinical Neurophysiology Society's (ACNS) guidelines determines electrocerebral inactivity (ECI). Studies have shown that ECI corroborated a clinical determination of brain death in 96.5% of adults but similar findings have not been reproduced in children [2,3,4].

Electrocerebral inactivity (previously called electrocerebral silence) is defined as the absence of any nonartifactual (cerebral) activity over 2 μ V (peak to peak) on a scalp EEG recorded using electrode pairs at least 10 cm apart and other technical criteria outlined by the ACNS.

ACNS Guidelines on minimal technical standards for EEG recording in suspected cases of cerebral death include the following recommendations [2]:

1. There should be a complete complement of scalp electrodes (including midline channels Fz, Cz, Pz which are relatively free of artifact and therefore useful to detect low-voltage cerebral activity).
2. Interelectrode impedances should be between 100 and 10,000 ohms to avoid distortion and amplification of extraneous signals.
3. The integrity of the entire recording system should be tested manually by touching each electrode with a pencil point or cotton swab to elicit artifact and verify the integrity of connections.
4. Electrode pairs should be at least 10 cm apart. Typically, the international 10–20 system has average adult interelectrode distances of 6 to 6.5 cm but increasing the interelectrode distance may show cerebral potentials otherwise not seen with usual sensitivities. The 10–10 system may also be used in addition to regularly used montages such as a central reference (Cz).
5. Use a sensitivity of 2 μ V/mm for at least 30 min with adequate and appropriate calibration. Thirty minutes is recommended as intermittent suppressions may mimic ECI.

6. High-frequency (low pass) filter should not be set below 30 Hz and low-frequency filter (high pass) should not be above 1 Hz to avoid attenuation of low-voltage fast activity or slow frequencies. Avoid using a 60 Hz notch filter.
7. Additional monitoring techniques such as EKG, EMG, and hand channels help identify artifactual activity. Continuous video recording aids interpretation.
8. The absence of reactivity to intense somatosensory, auditory, and visual stimuli should be confirmed. Electroretinograms can persist in response to photic stimulation.
9. Recordings should only be performed by a qualified technologist.
10. If inconclusive, the EEG should be repeated in about 6–24 hours.
11. Physiological variables such as temperature, blood pressure, oxygen saturation, and medications (especially sedatives and narcotics) should be considered when interpreting ECI. A toxicology screen may also be obtained.

Chapter Summary

1. The EEG is recommended as an ancillary test to confirm brain death in combination with a standardized neurological examination.
2. The EEG must be recorded using specific technical parameters outlined by the American Clinical Neurophysiology Society (ACNS) to determine electrocerebral inactivity (ECI) which highly correlates with brain death.
3. Electrocerebral inactivity is defined as the absence of any nonartifactual (cerebral) activity over 2 uV (peak to peak) on a scalp EEG recorded using electrode pairs at least 10 cm apart and other technical criteria outlined by the ACNS.

References

1. Russell JA, Epstein LG, Greer DM, et al. Brain death, the determination of brain death, and member guidance for brain death accommodation requests: AAN position statement. *Neurology*. 2019 Jan 29;**92**(5):228–32.
2. Stecker MM, Sabau D, Sullivan LR, et al. American Clinical Neurophysiology Society guideline 6: minimum technical standards for EEG recording in suspected cerebral death. *The Neurodiagnostic Journal*. 2016 Oct 1;**56**(4):276–84.
3. Fernández-Torre JL, Hernández-Hernández MA, Muñoz-Esteban C. Non-confirmatory electroencephalography in patients meeting clinical criteria for brain death: scenario and impact on organ donation. *Clinical Neurophysiology*. 2013 Dec 1;**124**(12):2362–7.
4. Nakagawa TA, Ashwal S, Mathur M, Mysore M, Society of Critical Care Medicine. Guidelines for the determination of brain death in infants and children: an update of the 1987 Task Force recommendations. *Pediatrics*. 2011 Sep;**128**(3).